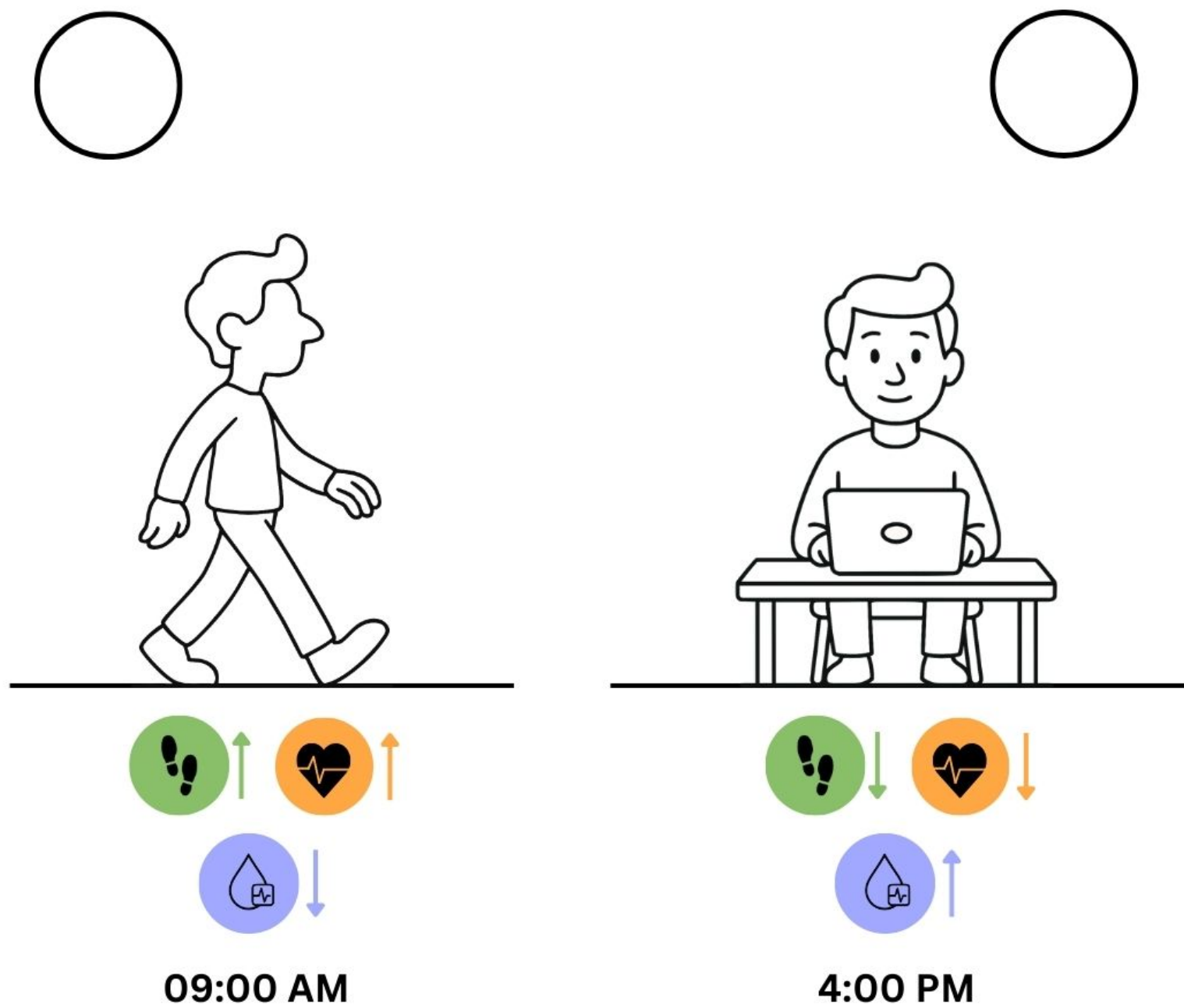


Learning Context-Aware Physiological Baselines from Wearable Data

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Overview



Disentangling stress-induced changes from context-driven variability is essential for accurate interpretation of HR data in real-world settings.

Research Questions

RQ1: Which contextual factors (e.g., activity, time of day, location) are associated with variations in perceived stress and physiological signals (e.g., HR)?

RQ2: Do context-aware physiological models that account for contextual confounds (e.g., physical activity, circadian rhythm) improve downstream stress classification performance compared to traditional methods?

Preliminary Results

Linear Mixed Model

Added predictor	LR statistic	Δdf	p-value
Steps (within-person z)	2265.42	1	$< 1e-300$
Time of Day	57.98	1	7.1E-06
Job vs Non-Job (JNJ)	12.71	1	3.6E-04
Location (home vs office)	0.36	1	0.55
Time demand (within-person z)	2.92	1	0.087
Mental demand (within-person z)	0.29	1	0.59

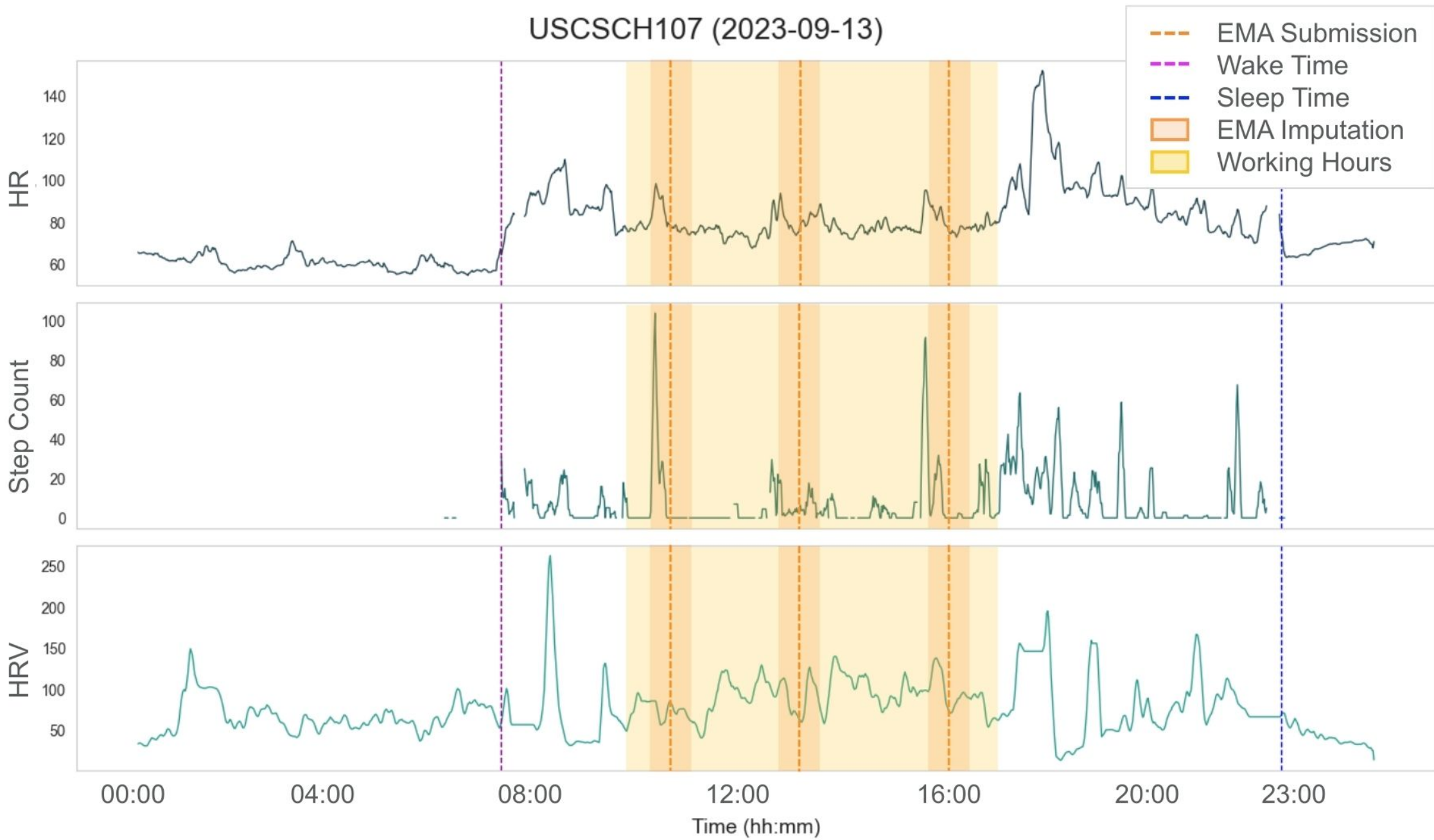
• 10-min bins, normalized

Downstream Task: Stress Classification

Model		F1
ar1_baseline_crhr_steps	0.50 ± 0.06	0.53 ± 0.15
ar1_gap_crhr	0.52 ± 0.06	0.55 ± 0.09
ar1_gap_crhr_steps	0.48 ± 0.06	0.56 ± 0.15
ar1_gap_crhr_steps_location	0.51 ± 0.04	0.60 ± 0.19
ar1_gap_steps	0.47 ± 0.04	0.56 ± 0.14
Raw bpm (no model)	0.48 ± 0.08	0.50 ± 0.06

Dataset

- Longitudinal data collection examining factors related to experiences of work-related stress over the course of a 4-month longitudinal data collection.
- Time-aligned physiological (Fitbit Charge 5), environmental, and work-related measures



Context-Integrated Modeling

- Dynamic linear models (**DLMs**) enable hierarchical statistical modeling of temporal dynamics and uncertainty [1].
- Markov Chain Monte Carlo (MCMC) can be used for Bayesian inference with a specified prior to infer posterior distributions over parameters.

$$\begin{aligned} y_t &= \mathbf{H}_t x_t + \mathbf{C}_t c_t + \varepsilon_t, & \varepsilon_t &\sim \mathcal{N}(0, \mathbf{R}_t) \\ x_t &= \mathbf{M}_t x_{t-1} + \eta_t, & \eta_t &\sim \mathcal{N}(0, \mathbf{Q}_t) \end{aligned} \quad [1] \quad c_t = \begin{cases} \text{Step Count} \\ \text{Circadian Rhythm} \\ \text{Location} \\ \text{Job Activity} \end{cases}$$

Where y_t is the Fitbit-observed HR
 x_t is the latent state

Discussion and Directions

Linear Mixed Model

- Activity dominates:** steps explain most HR variation
- Circadian Effects are present:** There exists HR variation associated with time-of-day that is not explained by activity
- Context matters:** Job vs NonJob context improves HR prediction beyond steps and time of day

Downstream Task: Stress Classification

- BDLM models not exhibiting large improvements over raw bpm on mini-subset of participants
- Utilize Job_NonJob context to boost model performance
- Further explore temporal alignment between EMAs and physiological deviations to improve interpretability and predictive validity of baselines.

Acknowledgements and References

This work is supported by the National Science Foundation under Grant No. IIS-2204942.

[1] Laine, M. (2019). Introduction to dynamic linear models for time series analysis. In *Geodetic time series analysis in earth sciences* (pp. 139-156). Cham: Springer International Publishing.

[2] Nazaret, A., Tonekaboni, S., Darnell, G., Ren, S. Y., Sapiro, G., & Miller, A. C. (2023). Modeling personalized heart rate response to exercise and environmental factors with wearables data. *NPJ Digital Medicine*, 6(1), 207.

[3] Rossi, A. D., Marzorati, D., Gerosa, T., Švihrová, R., Santini, S., & Faraci, F. (2025). Unobtrusive Perceived Sleep Quality Monitoring in the Wild. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 9(3), 1-26.

[4] Bowman, C., Huang, Y., Walch, O. J., Fang, Y., Frank, E., Tyler, J., ... & Forger, D. B. (2021). A method for characterizing daily physiology from widely used wearables. *Cell reports methods*, 1(4).